Semester Project

1. Project Overview

This semester project is designed to simulate a real-world **software house environment**. Students will work in teams and follow a complete **Software Development Life Cycle (SDLC)** including requirement gathering, design, development, testing, and project management.

Each team will independently select a project of their choice (e.g., web application, mobile app, management system, etc.), and execute it using industry practices and tools.

2. Team Formation

- Students must form **groups of 4–5 members**
- Each group must assign **clearly defined roles**

Mandatory Roles (Minimum 5)

Each team must include the following roles:

1. **Project Manager (PM)**
    - Responsible for planning, coordination, and timeline management
    - Ensures tasks are assigned and completed on time
2. **Business Analyst (BA) / Requirement Engineer**
    - Responsible for requirement gathering and documentation
    - Prepares SRS document
3. **UI/UX Designer (optional but recommended)**
    - Designs wireframes, mockups, and user interface
4. **Developer(s)**
    - Responsible for coding and implementation
    - At least 1–2 members
5. **QA Engineer / Tester**
    - Responsible for testing, bug reporting, and quality assurance
    - Prepares test plan and executes testing

Note: One student may handle more than one role only if the team size is 4.

3. Tools and Platforms (Mandatory Usage)

Each team must use the following tools:

- **JIRA** – for issue tracking, sprint planning, bug reporting
- OR **Trello** – for task management (if JIRA is not used)
- **GitHub** – for code version control
- Documentation tools (MS Word / Google Docs,overleaf)

All communication, task assignment, and progress tracking must be visible on the selected platform.

4. Project Execution Phases

The project must be completed in the following structured phases:

Phase 1: Project Selection & Proposal

- Select a project idea
- Define:
    - Problem statement
    - Objectives
    - Target users
- Submit a **Project Proposal Document**

Phase 2: Requirement Gathering & Analysis

- Identify:
    - Functional Requirements
    - Non-Functional Requirements
- Create:
    - Use Case Diagrams
    - User Stories (to be added in JIRA/Trello)

Deliverable:

- Complete **Software Requirements Specification (SRS)** document

Phase 3: Project Planning

- Create:
    - Project timeline (Gantt Chart or Sprint Plan)
    - Task breakdown (Work Breakdown Structure)
- Assign tasks in **JIRA/Trello**
- Define:
    - Milestones

    - Deadlines

Deliverable:

- Project Plan Document
- Active JIRA/Trello Board

Phase 4: System Design

- Create:
    - System Architecture Diagram
    - Database Design (ERD)
    - UI Wireframes / Mockups

Deliverable:

- Design Document

Phase 5: Development

- Implement the system in modules
- Follow proper coding practices
- Maintain code in GitHub with commits history

Important Requirements:

- Tasks must be linked with JIRA/Trello
- Each feature should correspond to a ticket

Phase 6: Testing & Quality Assurance

6.1 Test Planning

- Create a **Test Plan Document** including:
    - Testing scope
    - Testing types
    - Test strategy
    - Tools used

6.2 Test Case Design

- Write:

    - Test Cases
    - Test Scenarios

6.3 Test Execution

- Perform:
    - Functional Testing
    - Black Box Testing
    - Basic White Box concepts (if applicable)

6.4 Bug Reporting

- Report bugs on **JIRA**
    - Include:
        - Steps to reproduce
        - Expected vs actual result
        - Severity/Priority

Phase 7: Bug Fixing & Tracking

- Developers will:
    - Fix reported bugs
    - Update status on JIRA
- QA will:
    - Retest resolved issues
    - Close or reopen tickets

Deliverable:

- Bug Report Log (from JIRA/Trello)

Phase 8: Automation Testing (Basic Level)

- Perform basic automation using any tool (optional tools):
    - Selenium / Cypress / any scripting
- Automate at least:
    - 2–3 test cases

Deliverable:

- Automation scripts + brief explanation

Phase 9: Final Deployment & Demonstration

- Deploy the project (if possible)
- Prepare:
    - Final Presentation
    - System Demonstration

5. Required Deliverables

Each group must submit:

1. Project Proposal
2. SRS Document
3. Project Plan
4. Design Document
5. Source Code (GitHub link)
6. Test Plan Document
7. Test Cases & Test Results
8. Bug Reports (JIRA/Trello)
9. Automation Scripts
10. Final Presentation

6. Evaluation Criteria

Evaluation will be based on:

- Proper role distribution and teamwork
- Effective use of JIRA/Trello
- Quality of documentation (SRS, Test Plan)
- Implementation completeness
- Testing quality and bug tracking
- Use of automation
- Final demonstration

7. Important Instructions

- All team members must actively participate
- Work must be properly divided and traceable
- No copied projects are allowed
- Regular progress monitoring will be conducted
- JIRA/Trello activity will be evaluated as part of grading

8. Objective of the Project

The main goal is to ensure students:

- Understand real-world software development workflows
- Learn collaboration and role-based responsibilities
- Gain hands-on experience with industry tools
- Apply testing, documentation, and project management practices

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9. Evaluation Rubrics (Total: 100 Marks)

1. Team Roles & Collaboration (10 Marks)

Criteria	Marks	Description
Role Assignment	3	Clear definition of roles (PM, BA, Dev, QA, etc.)
Participation	4	Equal contribution from all members
Coordination	3	Effective communication and teamwork

2. Project Proposal (5 Marks)

Criteria	Marks	Description
Problem Definition	2	Clear and relevant problem statement
Objectives & Scope	3	Well-defined goals and feasibility

3. SRS Document (15 Marks)

Criteria	Marks	Description
Functional Requirements	5	Complete and clearly written
Non-Functional Requirements	3	Performance, usability, security, etc.
Use Cases / User Stories	4	Properly defined and structured
Documentation Quality	3	Clarity, formatting, completeness

4. Project Planning & Management (10 Marks)

Criteria	Marks	Description
Task Breakdown	3	Proper division of work
Timeline / Milestones	3	Realistic deadlines and planning
JIRA/Trello Usage	4	Active task tracking and updates

5. System Design (10 Marks)

Criteria	Marks	Description
Architecture Design	3	Logical and scalable design
Database Design (ERD)	3	Proper relationships and structure
UI/UX Design	4	Clear wireframes/mockups

6. Development & Implementation (15 Marks)

Criteria	Marks	Description
Functionality	6	Features implemented correctly
Code Quality	4	Clean, readable, modular code
Version Control	3	Proper use of GitHub commits
Integration	2	Modules work together properly

7. Testing & QA (15 Marks)

Criteria	Marks	Description
Test Plan	4	Well-structured and complete
Test Cases	4	Proper coverage and clarity
Testing Execution	4	Evidence of testing performed
Bug Identification	3	Ability to find relevant defects

8. Bug Tracking & Resolution (10 Marks)

Criteria	Marks	Description
Bug Reporting (JIRA)	4	Clear, detailed bug reports
Bug Resolution	4	Effective fixing of issues
Retesting	2	Proper validation of fixes

9. Automation Testing (5 Marks)

Criteria	Marks	Description
Automation Implementation	3	At least 2–3 test cases automated
Script Quality	2	Working and understandable scripts

10. Final Presentation & Demonstration (5 Marks)

Criteria	Marks	Description
Presentation Skills	2	Clear explanation of project
System Demo	3	Working and properly demonstrated system